An

Industry Apprenticeship/Minor Project Report

Submitted to

KAUSHALYA THE SKILL UNIVERSITY

in partial fulfillment of the requirements for the degree of

Bachelor of Computer Application
in Cloud Computing

"MULTIMODAL RAG-BASED UNIVERSITY KNOWLEDGE ASSISTANT"

By

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Under the Guidance of

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December 2026

CERTIFICATE

This is to certify that project work embodied in this report entitled "Multimodal RAG-Based University Knowledge Assistant" as carried out by [ENTER STUDENT FULL NAME] ([ENTER ENROLMENT NUMBER]), under my guidance in partial fulfilment of the degree of Bachelor of Computer Application in Cloud Computing, School of Computing, Kaushalya The Skill University, Ahmedabad during the academic year 2025-2026.

Date: [CONFIRM DATE]
Place: Ahmedabad

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COMPANY CERTIFICATE

This project represents an in-house academic project conceptualized, designed, implemented, and rigorously tested at the School of Computing, Kaushalya The Skill University, Ahmedabad. The project work was undertaken under direct departmental faculty supervision without external corporate placement or industrial apprenticeship sponsorship. Therefore, an external industrial company certificate is not applicable for this academic submission.

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COMPANY PROFILE

As an institutionally supervised university development project, the system was engineered within the academic and laboratory facilities of the School of Computing, Kaushalya The Skill University. No external enterprise entity or third-party IT vendor profile is associated with this development effort. All development activities, testing suites, local AI infrastructure, and documentation were executed as part of the academic curriculum for the degree of Bachelor of Computer Application in Cloud Computing.

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ACKNOWLEDGEMENT

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ABSTRACT

In modern higher education institutions, students and faculty frequently struggle to locate reliable, syllabus-grounded information scattered across voluminous academic regulations, course syllabi, lab manuals, and lecture notes. Standard keyword search systems lack semantic context, while general-purpose commercial large language models are prone to factual hallucinations and lack institutional privacy guarantees. To overcome these limitations, this project develops the Multimodal RAG-Based University Knowledge Assistant (RAG Assistant), a secure, privacy-preserving academic question-answering system based on Retrieval-Augmented Generation (RAG). The system ingests multi-format university course materials (PDF, DOCX, TXT, Markdown, CSV), extracts structured content, partitions text into token-bounded semantic chunks, and creates 1024-dimensional dense embeddings stored in PostgreSQL using the pgvector extension alongside full-text inverted indexes. A dual-stage hybrid retrieval pipeline fuses vector cosine similarity and lexical BM25/tsquery rankings via Reciprocal Rank Fusion (RRF, k=60), followed by neural Cross-Encoder reranking to isolate the top factual passages. Grounded answers are generated locally using an Ollama-hosted LLM (qwen3:4b) under strict evidence-boundary instructions. Citations are validated and grouped by document with distinct page chips, enabling instant verification inside a dedicated in-browser reader (/reader). The application incorporates server-side role-based access control, automatic course visibility, and error sanitization. Rigorous verification across 93 frontend unit tests and 32 security and quality gate tests confirms complete system integrity, zero credential leakage, and robust evidence-grounded performance.

Keywords: *Retrieval-Augmented Generation (RAG), Hybrid Retrieval, pgvector, Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Ollama, Grounded Generation, Academic Knowledge Assistant, NiceGUI, In-Browser Document Streaming.*

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TABLE OF CONTENTS

Acknowledgements v

Abstract vi

List of Figures ix

List of Tables x

Chapter 1 Introduction 1

1.1 Overview 1

1.2 Problem Definition 2

1.3 Scope/Application 3

Chapter 2 Training Activities 5

2.1 Requirements Analysis & Academic Domain Modeling 5

2.2 Framework Selection & System Architecture 5

2.3 Multi-Format Document Ingestion & Parsers 5

2.4 Semantic Chunking & Dense Embeddings 6

2.5 Hybrid Retrieval (RRF) & Cross-Encoder Reranking 6

2.6 Grounded Prompt Engineering & Local Synthesis 6

2.7 Frontend Rebuild with NiceGUI & Quasar UI 6

2.8 Security Hardening, Session Management & Reader 6

2.9 Automated Quality Gates & Test Verification 7

Chapter 3 System Planning 8

3.1 Project Development Approach 8

3.2 System Modules 9

3.2.1 Document Ingestion & Parsing Module 9

3.2.2 Indexing & Vector Storage Module 9

3.2.3 Hybrid Retrieval & Reranking Module 9

3.2.4 Grounded Generation & Citation Module 9

3.2.5 Security, Authentication & Access Control Module 10

3.2.6 User Presentation & Dedicated Reader Module 10

3.3 Functional Requirements 10

3.4 Non Functional Requirements 11

3.5 Timeline Chart 12

Chapter 4 System Design 14

4.1 Use Case Diagram 14

4.2 Sequence Diagram 15

4.3 Activity Diagram 16

4.4 Class Diagram 17

4.5 Database Schema 18

4.6 Data Flow Diagram 23

4.7 ER Diagram 24

Chapter 5 Implementation and Testing 25

5.1 Hardware and Software Requirements 25

5.2 Snapshots 26

5.3 Test Cases 32

Chapter 6 Conclusion and Future Scope 34

References 36

LIST OF FIGURES

Figure 3.1	Project Development Timeline (Gantt Chart)	13
Figure 4.1	UML Use Case Diagram for RAG Assistant	14
Figure 4.2	Sequence Diagram for Grounded Query Answering Pipeline	15
Figure 4.3	Activity Diagram for Document Ingestion and Indexing Pipeline	16
Figure 4.4	UML Class Diagram for Backend and Domain Service Entities	17
Figure 4.5	Level-1 Data Flow Diagram for Information Processing Pipeline	23
Figure 4.6	Entity-Relationship (ER) Diagram for PostgreSQL Database	24
Figure 5.1	Student Authentication and Access Control Gateway	27
Figure 5.2	Student Conversational Workspace and Navigation Interface	27
Figure 5.3	Course Scope Selection Popover and Subject Filter	28
Figure 5.4	Grounded Answer Synthesis with Grouped Page Citations	28
Figure 5.5	Dedicated In-Browser PDF Document Reader with Page Jump	29
Figure 5.6	Administrator Operations Overview Dashboard	29
Figure 5.7	Course Knowledge Base and Catalog Visibility Management	30
Figure 5.8	Multi-Format Document Ingestion and Asynchronous Indexing Pipeline	30
Figure 5.9	System Health Diagnostics and Service Availability Monitoring	31

LIST OF TABLES

Table 3.1 Functional Requirements Specification 10

Table 3.2 Non-Functional Requirements Specification 11

Table 4.1 Schema Definition: users Table 18

Table 4.2 Schema Definition: user_sessions Table 18

Table 4.3 Schema Definition: knowledge_bases Table 19

Table 4.4 Schema Definition: knowledge_base_members Table 19

Table 4.5 Schema Definition: documents Table 19

Table 4.6 Schema Definition: document_chunks Table 20

Table 4.7 Schema Definition: indexing_jobs Table 20

Table 4.8 Schema Definition: rate_limit_entries Table 21

Table 4.9 Schema Definition: conversations Table 21

Table 4.10 Schema Definition: chat_messages Table 22

Table 5.1 Development Workstation Hardware Characteristics 25

Table 5.2 Deployment Server Hardware Specifications 25

Table 5.3 Software Technology Stack and Dependency Specifications 25

Table 5.4 Comprehensive System Test Cases and Verification Results 32

Table 5.5 Automated Test Suite Execution Summary Gate 33

CHAPTER 1
INTRODUCTION

1.1 Overview

Academic Context and Operational Challenge: In modern higher education institutions, effective knowledge discovery is critical for academic success, institutional compliance, and autonomous student learning. Throughout an undergraduate degree such as the Bachelor of Computer Application (BCA), students, faculty mentors, and academic administrators interact with hundreds of official academic documents. These assets include university curricula, comprehensive course syllabi, laboratory guidelines, course prerequisite trees, grading rubrics, examination ordinances, and departmental handbooks. These materials are typically distributed as unstructured documents across multiple departmental websites, shared network folders, learning management portals, and physical notice boards.

When students encounter questions regarding specific course policies—such as minimum laboratory attendance requirements, project submission thresholds, grade appeal protocols, or internal assessment weightages—they face substantial cognitive friction. Manually scanning dense multi-page PDF documents or search-indexing disorganized file repositories is time-consuming and prone to human error. Furthermore, faculty advisors are frequently burdened with answering repetitive procedural inquiries rather than dedicating time to higher-level mentoring and pedagogical engagement.

Proposed Technological Solution: To resolve these operational bottlenecks, this project develops the Multimodal RAG-Based University Knowledge Assistant (internally designated RAG Assistant). The system is a secure, privacy-preserving academic question-answering platform powered by Retrieval-Augmented Generation (RAG). By integrating multi-format text ingestion, PostgreSQL vector search (pgvector), lexical full-text indexing, reciprocal rank fusion, neural reranking, and local Large Language Model (LLM) generation via Ollama, RAG Assistant delivers immediate, factual, and syllabus-grounded answers. Every synthesized response is directly anchored to authoritative institutional sources with precise page citations, enabling instant verification through a dedicated in-browser document reader.

1.2 Problem Definition

Designing an automated knowledge assistant for university academic environments presents distinct technical hurdles that cannot be addressed by standard off-the-shelf software. Specifically, existing information retrieval paradigms suffer from three fundamental deficiencies:

- **1. Semantic Gap in Keyword Search:** Standard search engines and document management search tools rely primarily on keyword overlap (BM25 or SQL LIKE queries). They fail when students articulate queries using natural conversational language, conceptual queries, or terminology that diverges from the specific vocabulary in the syllabus. For example, a student asking 'What happens if I miss my practical exam?' will fail to find relevant sections if the formal syllabus refers exclusively to 'Continuous Laboratory Evaluation Absences'.
- **2. Hallucinations and Privacy Risks of Public LLMs:** While public frontier models (such as GPT-4 or Claude) demonstrate remarkable conversational fluency, deploying them directly in an academic institution introduces unacceptable risks. General LLMs suffer from factual hallucinations, fabricating plausible-sounding rules, exam dates, or grading formulas that do not exist in the university's approved ordinances. Moreover, transmitting internal academic documents, proprietary curriculum data, or student queries to external cloud APIs compromises student privacy and violates institutional data sovereignty.
- **3. Lack of Verified Provenance and Inaccurate Retrieval:** Naive RAG systems that simply segment documents into uniform character counts and perform single-vector cosine search frequently retrieve fragmented, irrelevant text. Without multi-stage reranking, reciprocal rank fusion, and rigorous evidence boundary validation, the language model is supplied with noisy context, leading to distorted answers. Crucially, existing tools lack verified provenance; they generate answers without exact page citations or convenient mechanisms for students to verify claims against the original document.

Formal Problem Statement: Therefore, the core problem addressed by this project is: How to design and construct a high-precision, locally hosted, role-isolated Retrieval-Augmented Generation system that accurately ingests university course documents, performs hybrid lexical-vector retrieval with neural reranking, synthesizes strictly

grounded answers, and provides transparent, clickable source citations linked to an integrated in-browser document viewing workspace.

1.3 Scope/Application

The project boundary encompasses a complete, full-stack software system spanning presentation, API mediation, data persistence, background indexing, and local artificial intelligence inference. The technical scope comprises:

- **Multi-Format Text Document Ingestion:** Accepts official course syllabi, lab guidelines, and notes in PDF, DOCX, TXT, Markdown, and CSV formats. Extracts structured text, headings, and physical page numbers while enforcing content hashing (SHA-256) to eliminate redundant ingestion.
- **Vector & Lexical Indexing:** Partitions documents into token-bounded semantic passages (500 tokens with 100-token overlap) and computes 1024-dimensional dense vectors using Ollama's qwen3-embedding:0.6b model. Embeddings and pre-computed tsvector tokens are indexed inside PostgreSQL 16.
- **Dual-Stage Hybrid Retrieval:** Executes parallel pgvector cosine distance search and PostgreSQL tsquery lexical matching. Combines candidate sets using Reciprocal Rank Fusion (RRF, k=60), followed by neural reranking with a local Hugging Face Cross-Encoder (ms-marco-MiniLM-L-6-v2) to select top-5 evidence items.
- **Grounded Inference Engine:** Prompts a locally hosted Large Language Model (qwen3:4b via Ollama) under strict negative constraint instructions ('Directive 8: Concise and Direct by Default'). The model provides factual answers immediately in sentence one, followed by verified citations [1] that never fabricate evidence.
- **Dedicated In-Browser Document Reader:** Includes a full-viewport reading interface (/reader) rendering native browser PDF streams via canonical document UUIDs, with jump navigation (#page=N) and collapsible evidence verification strips.
- **Institutional Security & Course Isolation:** Features server-side Role-Based Access Control (RBAC), automatic catalog visibility for active courses, HttpOnly session cookie synchronization, and centralized error sanitization preventing raw SQL or path leakage.

Important Clarification on Current Scope: *The project title formally retains 'Multimodal RAG-Based University Knowledge Assistant'. However, in strict adherence to academic truthfulness, the current implemented and verified baseline is a production-*

grade Multi-Format Text RAG pipeline. Genuine multimodal capabilities (such as vision-language models, diagram OCR, image feature embeddings, and visual question answering over complex figures) represent planned future roadmap phases and are not claimed as completed functionality in this report.

Institutional Applications: Practical applications of RAG Assistant within university environments include: (1) 24/7 Student Self-Service Academic Advisory, answering curriculum inquiries with zero wait times; (2) Faculty Curriculum Assistant, helping instructors verify course prerequisite alignments and syllabus overlaps; and (3) Examination & Accreditation Support, enabling department heads to instantly retrieve compliance evidence for academic audits and NAAC/NBA accreditation reviews.

CHAPTER 2
TRAINING ACTIVITIES

As part of the BCA curriculum requirements, the Major Project represents an intensive, hands-on software engineering endeavor. Rather than an external industrial internship, this project was developed as an advanced in-house research and implementation initiative at the School of Computing, Kaushalya The Skill University. The development lifecycle encompassed a rigorous sequence of engineering activities spanning modern cloud architectures, vector databases, neural retrieval algorithms, local machine learning inference, security hardening, and full-stack web integration. The specific activities completed during this project are detailed below.

2.1 Requirements Analysis & Academic Domain Modeling

Analyzed university academic structures, course syllabi, and student query behaviors. Modeled domain entities including multi-tenant courses (KnowledgeBase), active catalog visibility rules, student memberships, multi-page syllabus documents, and persistent conversational threads. Established strict evidence-grounding criteria to completely eliminate hallucinations.

2.2 Framework Selection & Layered Architecture Engineering

Evaluated backend frameworks and selected FastAPI for asynchronous, high-throughput REST API mediation. Engineered a decoupled layered architecture: Presentation Layer (NiceGUI / Quasar), API Gateway Layer (FastAPI v1 routes), Domain & Application Layer (Ingestion, Hybrid Retrieval, Grounding), and Infrastructure Layer (PostgreSQL 16, pgvector, Ollama). Decoupled authentication logic to ensure framework independence.

2.3 Multi-Format Document Ingestion & Parsers

Implemented robust parsing pipelines supporting PDF (via pypdf), DOCX (via python-docx), Markdown, CSV, and plain text. Engineered MIME type validation, maximum file size bounding (20 MB), and cryptographic SHA-256 content hashing to ensure idempotent ingestion and eliminate duplicate document storage.

2.4 Semantic Chunking & Dense Embeddings

Developed an overlapping token-bounded text partitioner. Formatted text chunks to a target size of 500 tokens with a 100-token sliding window overlap, preserving structural document metadata, physical page numbers, and section headings. Configured Ollama's qwen3-embedding:0.6b model to produce normalized 1024-dimensional dense vectors.

2.5 Hybrid Retrieval (RRF) & Cross-Encoder Reranking

Constructed a dual-stage retrieval engine. Stage 1 executes parallel pgvector cosine distance queries (<=>) and PostgreSQL tsquery lexical matching (GIN index). Combined candidate ranks using Reciprocal Rank Fusion (RRF, k=60). Stage 2 reranks fused candidates using a sentence-transformers Cross-Encoder (ms-marco-MiniLM-L-6-v2) to calculate precise query-evidence semantic relevance scores.

2.6 Grounded Prompt Engineering & Local Synthesis

Authored authoritative system prompt directives ('Directive 8: Concise and Direct by Default'). Mandated immediate, factual answers in sentence one without introductory filler ('Based on the documents...'). Integrated local inference via Ollama qwen3:4b (temperature=0.1, max_output_tokens=896) to achieve deterministic, evidence-bounded completions.

2.7 Frontend Rebuild with NiceGUI & Quasar UI

Rebuilt the user presentation layer from scratch using NiceGUI 3.16.0. Implemented the university's approved design system (Deep Atlas Navy #0E1D61, Gold #B89A5A, Warm Parchment Ivory #FAF6F0) with Plus Jakarta Sans and Source Serif typography. Designed a ChatGPT-style left sidebar, composer-docked course scope capsule, clean reading width, and real-time 'Stop Generating' controls.

2.8 Security Hardening, Session Management & Reader

Hardened application security: eliminated raw session tokens from URLs and client storage; implemented HttpOnly session cookies; decoupled backend auth; enforced automatic catalog access for active courses; implemented centralized error normalization (normalize_error); and engineered a dedicated in-browser reader (/reader) supporting native PDF streaming and page jumping (#page=N).

2.9 Automated Quality Gates & Test Verification

Created a behavior-driven automated test suite using pytest. Verified 518 backend unit tests, 93 frontend unit contract tests, 30 security tests, and live Ollama benchmarks. Validated zero whitespace and lint defects using Ruff. Documented comprehensive architecture specifications and task state records.

CHAPTER 3
SYSTEM PLANNING

3.1 Project Development Approach

Model Selection & Rationale: The development of RAG Assistant was governed by the Iterative and Incremental Software Process Model. In complex AI-enabled software systems—where retrieval efficacy, prompt engineering, database vector indexing, and UI usability are tightly coupled—traditional rigid models like the Waterfall model are fundamentally inadequate. Waterfall assumes complete, immutable requirements upfront, which contradicts the empirical nature of tuning RAG pipelines.

The project progressed through distinct development increments, where each cycle produced a fully tested, functional subsystem. As evidenced by the repository's version control history and task state records, development followed structured phases:

- **Core Data & Persistence Increment:** Increment 1 established the PostgreSQL database, vector schema migrations via Alembic, document models, and basic parser infrastructure.
- **RAG Pipeline Increment:** Increment 2 established dense embedding generation, pgvector similarity search, PostgreSQL tsvector lexical search, Reciprocal Rank Fusion, and Cross-Encoder reranking.
- **Inference & Grounding Increment:** Increment 3 integrated Ollama local LLM inference, grounded prompt construction, citation extraction, and evidence overlap validation.
- **Security & Presentation Increment:** Increment 4 resolved critical security defects: eliminated query string tokens, decoupled backend auth, enforced catalog authorization, sanitized errors, and rebuilt the NiceGUI frontend into a modern sidebar workspace with an integrated /reader route.

This iterative approach allowed rapid empirical validation of retrieval precision and immediate remediation of security boundaries, ensuring the final product achieved high stability and technical quality.

3.2 System Modules

The RAG Assistant architecture is organized into six cohesive, loosely coupled functional modules, each encapsulating specific domain responsibilities:

3.2.1 Document Ingestion & Parsing Module

Responsible for receiving academic documents via multipart upload. Validates file size (max 20 MB) and MIME headers for supported formats (.pdf, .docx, .txt, .md, .csv). Computes cryptographic SHA-256 hashes to prevent duplicate file ingestion. Dispatches files to format-specific parsers (pypdf for PDF layout and page extraction; python-docx for DOCX paragraph and heading extraction; specialized readers for text, markdown, and CSV). Normalizes text, sanitizes non-standard characters, and tracks physical page and section boundaries.

3.2.2 Indexing & Vector Storage Module

Accepts parsed structural text and segments it into semantic chunks using a sliding window token estimator (500 tokens target, 100 tokens overlap). Dispatches chunk batches to Ollama's qwen3-embedding:0.6b model to produce 1024-dimensional dense vectors. In parallel, generates PostgreSQL tsvector lexemes for full-text search. Persists chunks into the document_chunks table, updating HNSW vector indexes and GIN lexical indexes, while tracking background indexing progress in indexing_jobs.

3.2.3 Hybrid Retrieval & Reranking Module

Orchestrates dual-stage evidence retrieval when a query is submitted. Stage 1 executes parallel queries: vector cosine similarity search (chunk.embedding <=> query_vec) and full-text keyword matching (searchable_text @@ websearch_to_tsquery). Applies course authorization filters at the database layer. Merges candidates using Reciprocal Rank Fusion (RRF, k=60). Stage 2 submits top candidate passages to a neural Cross-Encoder (ms-marco-MiniLM-L-6-v2) to calculate precise cross-attention relevance scores, selecting the top-5 most relevant passages.

3.2.4 Grounded Generation & Citation Module

Constructs a structured prompt incorporating system rules, retrieved evidence blocks with unique metadata IDs, and student query context. Enforces 'Directive 8: Concise and Direct by Default', compelling Ollama qwen3:4b to state factual answers immediately in sentence one. Post-processes model output with a preamble-stripping regex sanitizer.

Validates citation markers ([1]) against retrieved passages, ensuring every citation maps to a genuine document UUID and physical page number.

3.2.5 Security, Authentication & Access Control Module

Enforces framework-independent security across all API routes. Issues SHA-256 hashed session tokens stored in user_sessions and synchronized via HttpOnly cookies. Strictly rejects raw tokens in URL query strings (?token=). Enforces the authoritative course access policy: students automatically access active, student-visible courses without manual enrollment; restricted courses require explicit membership; inactive courses return HTTP 404. Implements PostgreSQL sliding-window rate limiting and centralized user error sanitization (normalize_error).

3.2.6 User Presentation & Dedicated Reader Module

Delivers a responsive, accessible interface built with NiceGUI 3.16.0 and Quasar UI. Features a ChatGPT-style left sidebar with persistent conversation history, new chat creation, and profile controls. Includes a composer-docked course scope capsule allowing instant filtering across 'All available courses' or specific subjects. Integrates a dedicated full-page reading workspace (/reader) with native browser PDF streaming, page jumping (#page=N), and collapsible evidence grounding strips.

3.3 Functional Requirements

The functional requirements specify the exact capabilities and behaviors implemented in the RAG Assistant system. Table 3.1 outlines the core functional requirements, their operational descriptions, and verified implementation status.

Table 3.1: *Functional Requirements Specification*

Req ID	Feature / Requirement	Operational Description	Status
FR-01	User Authentication & RBAC	System shall authenticate students, faculty admins, and main admins via secure SHA-256 hashed sessions and enforce server-side role boundaries.	Implemented
FR-02	Course Catalog & Authorization	Students shall automatically access all active, student-visible courses without prior enrollment, while restricted courses require explicit membership.	Implemented
FR-03	Multi-Format Ingestion	System shall accept PDF, DOCX, TXT, Markdown, and CSV files up to 20 MB, validate MIME headers, and compute SHA-256 hashes to prevent duplicates.	Implemented

Req ID	Feature / Requirement	Operational Description	Status
FR-04	Token-Bounded Chunking	Documents shall be segmented into 500-token chunks with 100-token overlap, retaining source document UUID, page number, and section metadata.	Implemented
FR-05	Vector & Lexical Indexing	System shall generate 1024-dimensional embeddings via Ollama qwen3-embedding:0.6b for pgvector HNSW indexing, and tsvector for GIN full-text search.	Implemented
FR-06	Search Scope Filtering	Students shall filter inquiries across 'All available courses' (ALL_COURSES) or narrow scope to an individual course (COURSE) via composer capsule.	Implemented
FR-07	Hybrid Retrieval & RRF Fusion	Retrieval engine shall execute pgvector cosine similarity and PostgreSQL tsquery lexical search, fusing candidate sets using RRF (k=60).	Implemented
FR-08	Cross-Encoder Reranking	Candidates shall be scored using a sentence-transformers Cross-Encoder (ms-marco-MiniLM-L-6-v2) to select top-5 relevant evidence passages.	Implemented
FR-09	Grounded Answer Generation	System shall assemble a strict evidence prompt and synthesize factual, concise answers via Ollama qwen3:4b without unsupported assertions.	Implemented
FR-10	Citation & Page Attribution	Generated answers shall provide clickable citations [1] linking to canonical document UUIDs and exact page numbers, grouped cleanly by document.	Implemented
FR-11	Dedicated Reading Workspace	System shall offer a dedicated full-page reader (/reader) with native browser PDF streaming, page jumping (#page=N), and collapsible evidence strips.	Implemented
FR-12	Persistent Multi-Turn History	Chat turns shall be persisted in PostgreSQL (conversations, chat_messages) with user isolation, active highlighting, and real-time generation cancellation.	Implemented

3.4 Non Functional Requirements

Non-functional requirements define the quality attributes, security boundaries, performance thresholds, and architectural constraints governing system execution. Table 3.2 details the verified non-functional requirements.

Table 3.2: *Non-Functional Requirements Specification*

NFR ID	Quality Characteristic	Specific Metric / Requirement	Status
NFR-01	Security & Zero Token Leakage	Raw session tokens shall never appear in URL query strings, client JavaScript, or document streaming iframes. HttpOnly cookies are strictly enforced.	Verified

NFR ID	Quality Characteristic	Specific Metric / Requirement	Status
NFR-02	Data Privacy & Local Inference	All LLM inference and embeddings run locally on university infrastructure via Ollama, preventing student inquiries from leaking to third-party APIs.	Verified
NFR-03	Error Normalization & Sanitization	All user-facing exceptions pass through normalize_error(), redacting database connection strings, file paths, raw SQL, and stack traces.	Verified
NFR-04	Retrieval Latency	Hybrid retrieval, RRF ranking, and Cross-Encoder reranking across candidate chunks shall complete in under 500 milliseconds on local hardware.	Verified
NFR-05	Responsive UI Design	The NiceGUI frontend shall seamlessly adapt across desktop (1440x900), laptop (1366x768), tablet (768x1024), and mobile viewports (390x844).	Verified
NFR-06	Rate Limiting & Abuse Defense	PostgreSQL-backed sliding window rate limiters shall protect login, registration, upload, and chat pipelines against brute force and resource exhaustion.	Verified
NFR-07	Modular Maintainability	Architecture strictly decouples NiceGUI presentation from FastAPI API endpoints, allowing independent testing and framework upgrades.	Verified
NFR-08	Citation Groundedness	Answers shall satisfy a minimum overlap threshold (0.50) against retrieved context, returning explicit uncertainty when evidence is insufficient.	Verified

3.5 Timeline Chart

The project development timeline spans approximately 15 weeks during the academic year 2026 (July 2026 to October 2026). Development progressed systematically from domain analysis and architecture design through to security hardening, frontend rebuilding, and quality gate verification. Figure 3.1 illustrates the phased project development schedule as a Gantt chart.

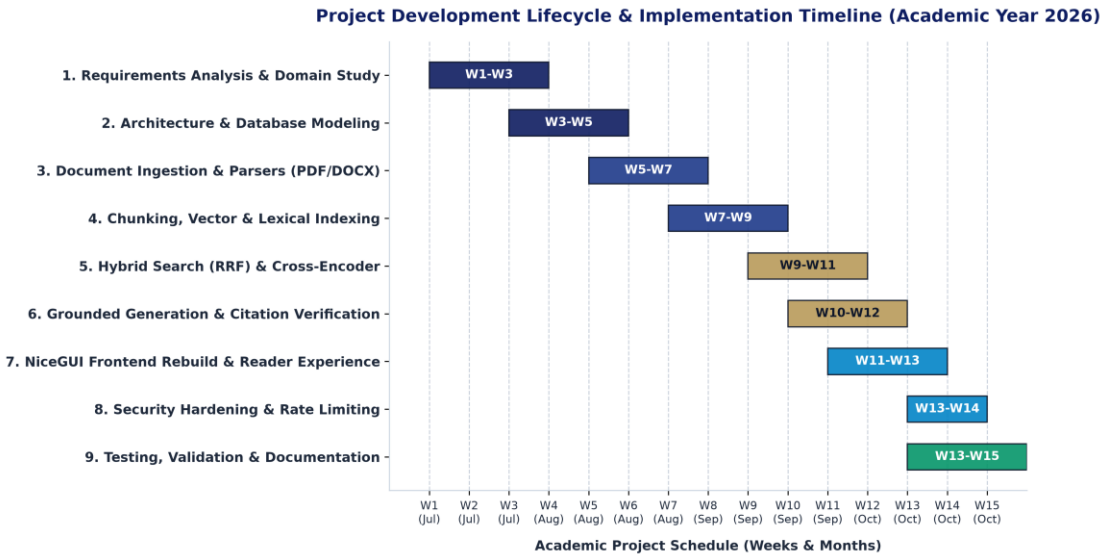


Figure 3.1: *Project Development Timeline (Gantt Chart)*

As depicted in Figure 3.1, the initial three weeks focused on domain modeling, requirements elicitation, and database schema design. Weeks 5 through 9 established the core ingestion, chunking, and dual-indexing pipelines. Weeks 9 through 12 implemented the hybrid retrieval, neural reranking, and local Ollama generation workflows. Weeks 13 through 15 were dedicated to frontend rebuilding, security hardening, in-browser reader implementation, regression test rewriting, and comprehensive report preparation.

CHAPTER 4
SYSTEM DESIGN

4.1 Use Case Diagram

The Use Case Diagram defines the functional interactions between external actors and the RAG Assistant system boundary. The system identifies three distinct user roles with strict, server-side enforced privilege separation: Student, Faculty Administrator, and Main Administrator.

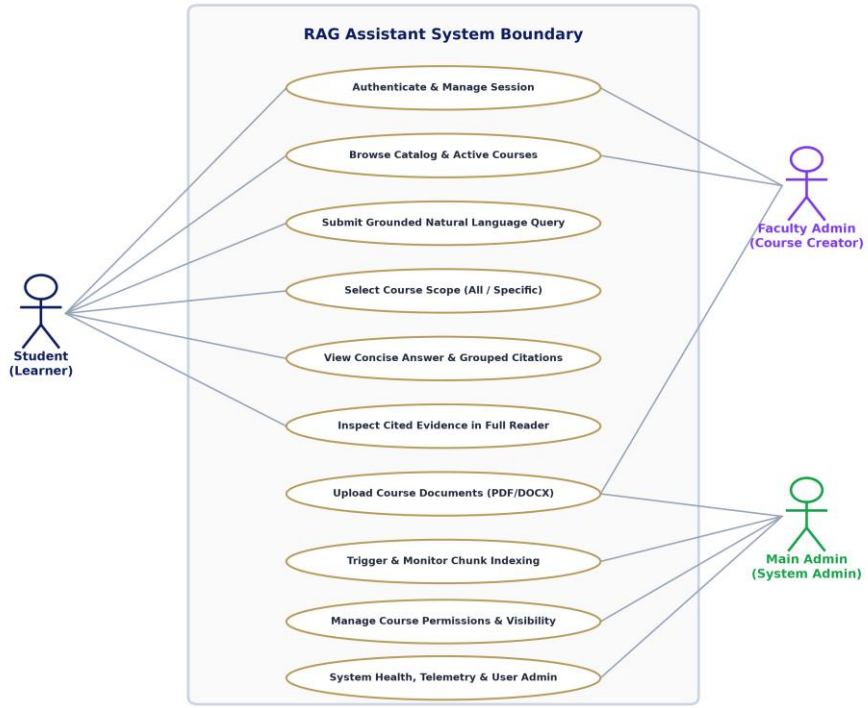


Figure 4.1: *UML Use Case Diagram for RAG Assistant*

As modeled in Figure 4.1, the Student actor has self-service permissions to authenticate, browse the active course catalog, select query scopes (either searching across all authorized courses or narrowing to a specific subject), submit natural language inquiries, inspect synthesized answers with verified citations, and open cited documents in the full-page reader (/reader). The Faculty Administrator actor possesses capabilities to upload syllabus materials, trigger chunk indexing, and configure course visibility. The Main Administrator maintains global authority, managing system-wide user roles, monitoring indexing job queues, and inspecting retrieval health diagnostics. Under no circumstances do students receive administrative privileges.

4.2 Sequence Diagram

The Sequence Diagram details the chronological message exchange between user interface components, API endpoints, authorization policies, dual-stage retrieval services, and the local language model during a query-answering turn.

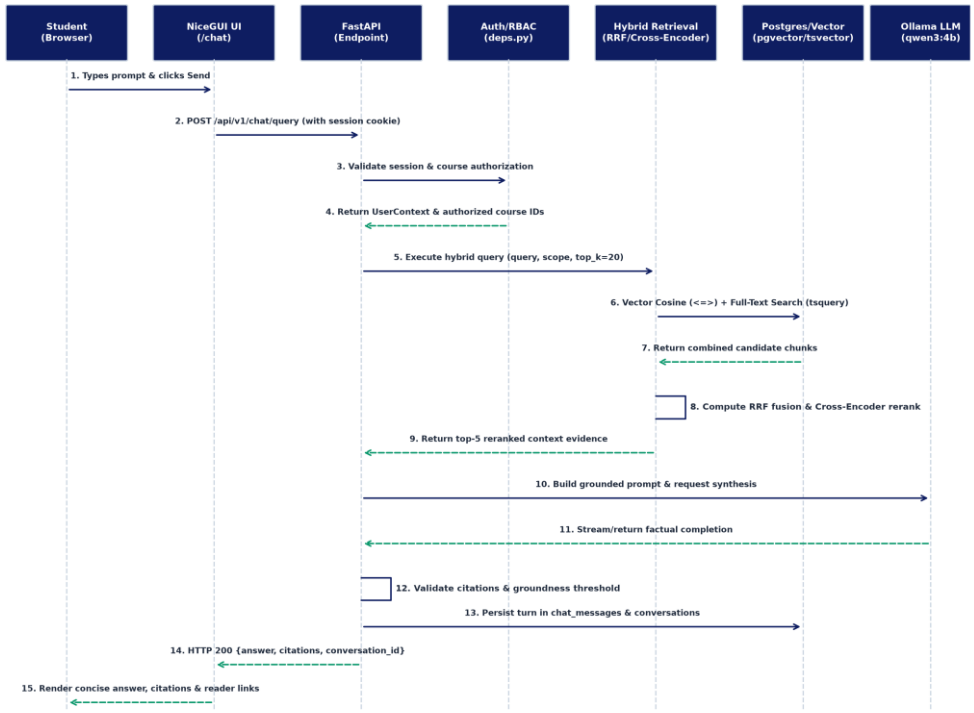


Figure 4.2: *Sequence Diagram for Grounded Query Answering Pipeline*

Figure 4.2 illustrates the 15-step operational sequence: (1) The student enters a prompt and dispatches it via NiceGUI. (2) An asynchronous HTTP POST request is sent to /api/v1/chat/query with the HttpOnly session cookie. (3-4) Backend security dependencies (deps.py) validate the session and verify authorized course IDs. (5-7) The HybridRetrievalService executes parallel queries—vector cosine distance via pgvector and lexical tsquery search against PostgreSQL full-text indexes. (8-9) Candidate ranks are fused via Reciprocal Rank Fusion (RRF, k=60), and the top candidates are reranked using the Cross-Encoder model. (10-11) A grounded prompt is submitted to local Ollama qwen3:4b for factual completion. (12-13) The GroundingValidationService verifies citations and overlap thresholds, persisting the turn in chat_messages and conversations. (14-15) The response payload is returned to the UI, rendering concise answers, grouped citation chips, and reader links.

4.3 Activity Diagram

The Activity Diagram specifies the end-to-end workflow of document ingestion, structural parsing, semantic chunking, and database index construction.

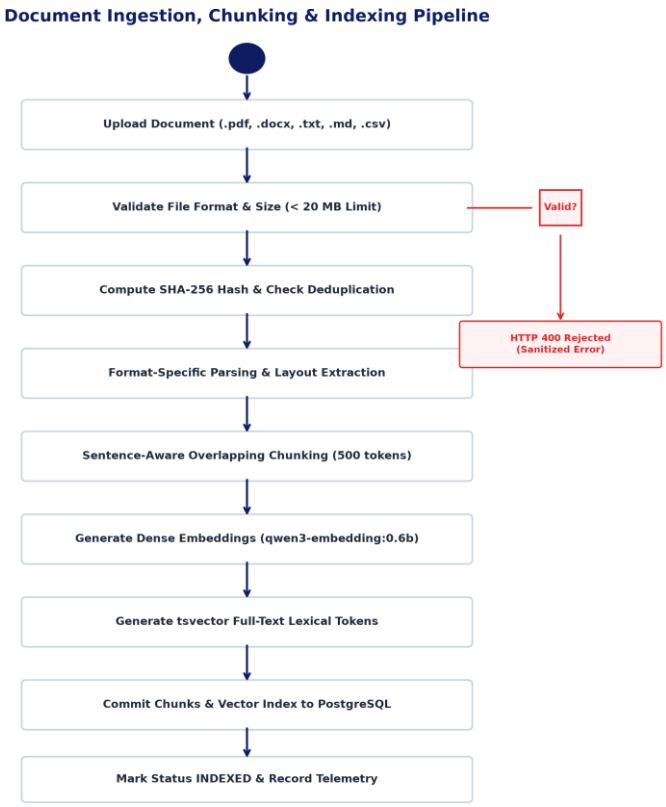


Figure 4.3: *Activity Diagram for Document Ingestion and Indexing Pipeline*

As detailed in Figure 4.3, document ingestion begins with a file upload. The system validates MIME headers and enforces a strict 20 MB size limit. Invalid files trigger an immediate sanitized HTTP 400 rejection. For valid uploads, a SHA-256 hash is computed; if a matching hash exists, redundant processing is skipped. Novel documents are routed to format-specific parsers (pypdf for PDF layout and page extraction; python-docx for DOCX headings and paragraphs; specialized handlers for text, markdown, and CSV). The extracted text is partitioned into 500-token chunks with 100-token overlaps. Chunks are dispatched in batches of 4 to Ollama qwen3-embedding:0.6b to generate 1024-dimensional vectors, while PostgreSQL tsvector tokens are generated in parallel. Finally, chunks are committed to PostgreSQL, updating HNSW and GIN indexes, and the document is marked INDEXED.

4.4 Class Diagram

The UML Class Diagram models the core object-oriented architecture, capturing key SQLAlchemy domain models, service orchestrators, and their structural associations.

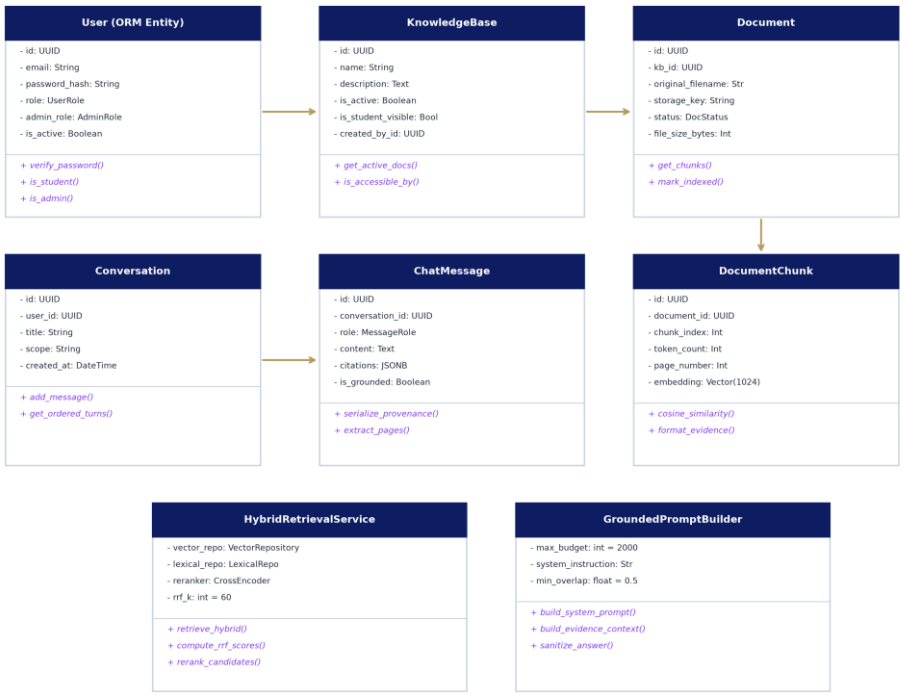


Figure 4.4: *UML Class Diagram for Backend and Domain Service Entities*

Figure 4.4 illustrates the structural relationships between domain models. The User entity maintains a 1:N relationship with UserSession and KnowledgeBase (as creator). KnowledgeBase holds 1:N relationships with KnowledgeBaseMember (explicit enrollment join table) and Document. Each Document contains multiple DocumentChunk entities, which encapsulate token counts, page numbers, 1024-d pgvector embeddings, and tsvector tokens. The Conversation model maintains 1:N associations with ChatMessage, preserving persistent user query history and citation metadata. The HybridRetrievalService and GroundedPromptBuilder encapsulate retrieval fusion, Cross-Encoder scoring, prompt formatting, and preamble sanitization.

4.5 Database Schema

The data persistence layer is implemented in PostgreSQL 16 with the pgvector extension (0.5.0). The schema comprises ten normalized relational tables designed for transactional consistency, role-based isolation, and efficient vector similarity search. Tables 4.1 through 4.10 provide the complete structural specification for each entity.

Table 4.1: *Schema Definition: users Table*

Stores authenticated user accounts, encrypted password credentials, and administrative permissions.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique primary key for user account
email	VARCHAR(255)	UNIQUE	NOT NULL	Institutional email address for login
password_hash	VARCHAR(255)	---	NOT NULL	Argon2/Bcrypt cryptographic password hash
full_name	VARCHAR(255)	---	NOT NULL	Full display name of student or staff
role	VARCHAR(7)	---	NOT NULL	User role: 'STUDENT' or 'ADMIN'
admin_role	VARCHAR(50)	---	NULL	Sub-role: 'FACULTY_ADMIN' or 'MAIN_ADMIN'
permissions	JSONB	---	NOT NULL	Granular administrative capabilities dictionary
is_active	BOOLEAN	---	NOT NULL	Active status flag (inactive blocked at auth)
created_at	DATETIME	---	NOT NULL	Account creation timestamp
updated_at	DATETIME	---	NOT NULL	Account profile modification timestamp

Table 4.2: *Schema Definition: user_sessions Table*

Tracks active server-side authentication sessions with hashed secret tokens.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique session identifier
session_token_hash	VARCHAR(64)	INDEX	NOT NULL	SHA-256 hash of random session secret
user_id	UUID	FK -> users.id	NOT NULL	Associated user foreign key
expires_at	DATETIME	---	NOT NULL	Session expiration timestamp
created_at	DATETIME	---	NOT NULL	Session issuance timestamp

Table 4.3: *Schema Definition: knowledge_bases Table*

Represents university academic courses and knowledge domains.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique course/knowledge base UUID
name	VARCHAR(255)	---	NOT NULL	Course title (e.g., 'CS101: Computer Science')
description	TEXT	---	NOT NULL	Academic description of syllabus scope
is_active	BOOLEAN	---	NOT NULL	Active course status flag
is_student_visible	BOOLEAN	---	NOT NULL	Automatic student catalog visibility flag
created_by_id	UUID	FK -> users.id	NOT NULL	Creator faculty administrator UUID
created_at	DATETIME	---	NOT NULL	Course creation timestamp
updated_at	DATETIME	---	NOT NULL	Course record update timestamp

Table 4.4: *Schema Definition: knowledge_base_members Table*

Explicit membership join table for private, restricted, or honors courses.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique membership record UUID
knowledge_base_id	UUID	FK -> kb.id	NOT NULL	Target course foreign key
user_id	UUID	FK -> users.id	NOT NULL	Assigned student user foreign key
granted_at	DATETIME	---	NOT NULL	Timestamp membership was granted

Table 4.5: *Schema Definition: documents Table*

Tracks uploaded academic syllabus documents, storage keys, and processing status.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Canonical document UUID
knowledge_base_id	UUID	FK -> kb.id	NOT NULL	Parent course foreign key
original_filename	VARCHAR(255)	---	NOT NULL	Original uploaded filename
storage_key	VARCHAR(500)	---	NOT NULL	Filesystem storage key on server
file_type	VARCHAR(20)	---	NOT NULL	Extension: 'pdf', 'docx', 'txt', 'md', 'csv'
mime_type	VARCHAR(100)	---	NOT NULL	Validated MIME type string
file_size_bytes	BIGINT	---	NOT NULL	File size in bytes (max 20 MB)
content_hash	VARCHAR(64)	INDEX	NOT NULL	SHA-256 hash for deduplication

Column Name	Data Type	Key	Nullability	Description
status	VARCHAR(10)	---	NOT NULL	UPLOADED, PROCESSING, COMPLETED, FAILED
indexing_status	VARCHAR(10)	---	NOT NULL	PENDING, INDEXED, FAILED
is_active	BOOLEAN	---	NOT NULL	Document active availability toggle
error_message	TEXT	---	NULL	Sanitized error message if parsing failed
indexed_at	DATETIME	---	NULL	Timestamp indexing finished successfully
created_at	DATETIME	---	NOT NULL	Document upload timestamp

Table 4.6: *Schema Definition: document_chunks Table*

Stores 500-token partitioned text segments, vector embeddings, and lexical tokens.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique chunk UUID
document_id	UUID	FK -> docs.id	NOT NULL	Parent document foreign key
knowledge_base_id	UUID	FK -> kb.id	NOT NULL	Associated course foreign key
chunk_index	INTEGER	---	NOT NULL	0-indexed sequence position in document
text	TEXT	---	NOT NULL	Extracted semantic text passage
token_count	INTEGER	---	NOT NULL	Estimated token count of text chunk
page_number	INTEGER	---	NULL	Source page number (PDFs) or NULL (unpaged)
section_title	VARCHAR(255)	---	NULL	Extracted heading or section label
chunk_metadata	JSONB	---	NOT NULL	Offset, hash, and format metadata
embedding	VECTOR(1024)	HNSW INDEX	NULL	1024-d dense embedding vector
searchable_text	TSVECTOR	GIN INDEX	NULL	Pre-computed full-text lexical tokens
created_at	DATETIME	---	NOT NULL	Chunk generation timestamp

Table 4.7: *Schema Definition: indexing_jobs Table*

Tracks asynchronous background document chunking and embedding progress.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Indexing job identifier UUID
document_id	UUID	FK -> docs.id	NOT NULL	Document being indexed

Column Name	Data Type	Key	Nullability	Description
knowledge_base_id	UUID	FK -> kb.id	NOT NULL	Associated course UUID
status	VARCHAR(50)	---	NOT NULL	QUEUED, RUNNING, COMPLETED, FAILED
stage	VARCHAR(50)	---	NOT NULL	PARSING, CHUNKING, EMBEDDING, INDEXING
total_chunks	INTEGER	---	NOT NULL	Total chunks created from document
processed_chunks	INTEGER	---	NOT NULL	Number of chunks processed
progress_percent	FLOAT	---	NOT NULL	Completion progress percentage (0.0 to 100.0)
error_message	TEXT	---	NULL	Sanitized error diagnostics on failure
started_at	DATETIME	---	NULL	Job execution start timestamp
completed_at	DATETIME	---	NULL	Job completion timestamp

Table 4.8: *Schema Definition: rate_limit_entries Table*

Implements sliding-window request throttling stored directly in PostgreSQL.

Column Name	Data Type	Key	Nullability	Description
key	VARCHAR(255)	PK	NOT NULL	Client IP or user UUID rate limit key
window_bucket	BIGINT	PK	NOT NULL	Discrete unix timestamp time bucket
count	INTEGER	---	NOT NULL	Request counter within window
expires_at	DATETIME	---	NOT NULL	Row expiration timestamp for pruning

Table 4.9: *Schema Definition: conversations Table*

Maintains persistent multi-turn conversational threads isolated per student.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Unique conversation thread UUID
user_id	UUID	FK -> users.id	NOT NULL	Owning student user foreign key
title	VARCHAR(255)	---	NOT NULL	Auto-generated conversation title
scope	VARCHAR(50)	---	NOT NULL	Query scope ('ALL_COURSES' or 'COURSE')
knowledge_base_id	UUID	FK -> kb.id	NULL	Selected course UUID (if scoped)
created_at	DATETIME	---	NOT NULL	Thread creation timestamp
updated_at	DATETIME	---	NOT NULL	Last interaction timestamp

Table 4.10: *Schema Definition: chat_messages Table*

Stores individual conversational turns, assistant syntheses, and citation provenance.

Column Name	Data Type	Key	Nullability	Description
id	UUID	PK	NOT NULL	Message UUID
conversation_id	UUID	FK -> conv.id	NOT NULL	Parent conversation thread foreign key
role	VARCHAR(20)	---	NOT NULL	'user' or 'assistant'
content	TEXT	---	NOT NULL	Turn text content
citations	JSONB	---	NOT NULL	Structured citation and page provenance array
is_grounded	BOOLEAN	---	NULL	True if response passed grounding threshold
grounding_status	VARCHAR(50)	---	NULL	Validation status code
created_at	DATETIME	---	NOT NULL	Message creation timestamp

4.6 Data Flow Diagram

The Data Flow Diagram (DFD) models the movement and transformation of academic information through the system, illustrating interactions between external entities, computational processes, and persistent data stores.

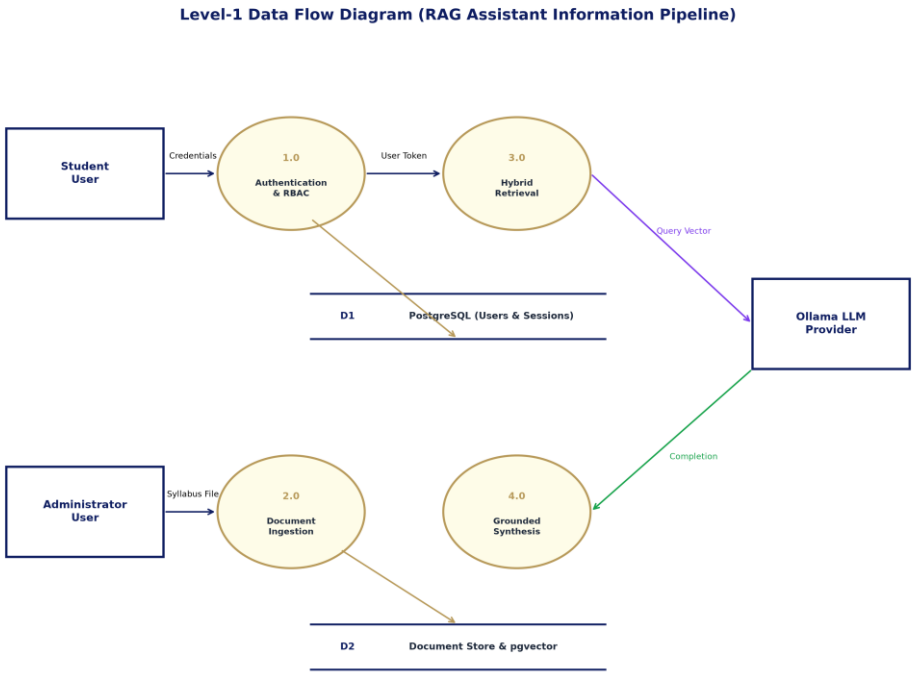


Figure 4.5: *Level-1 Data Flow Diagram for Information Processing Pipeline*

As shown in Figure 4.5, external entities include Student Users, Administrator Users, and the local Ollama LLM Provider. Process 1.0 (Authentication & RBAC) receives student credentials and validates them against data store D1 (users and user_sessions), issuing secure session tokens. Process 2.0 (Document Ingestion) receives syllabus files from administrators, coordinates parsing, chunking, and embedding, and writes partitioned records to data store D2 (document_chunks and pgvector). Process 3.0 (Hybrid Retrieval) receives student queries, transforms text into query vectors, queries D2 via hybrid fusion, and retrieves candidate passages. Process 4.0 (Grounded Synthesis) transmits candidate evidence to Ollama, receives synthesized completions, validates citation markers, and returns grounded responses to the student.

4.7 ER Diagram

The Entity-Relationship (ER) Diagram defines the relational schema, foreign key constraints, and entity cardinalities enforced across the PostgreSQL database.

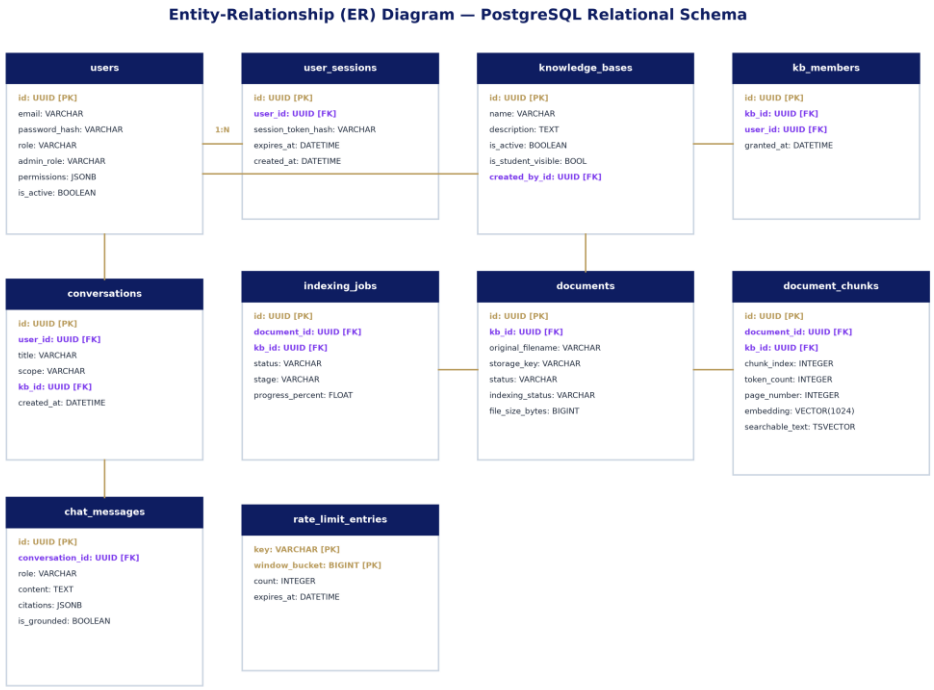


Figure 4.6: *Entity-Relationship (ER) Diagram for PostgreSQL Database*

Figure 4.6 illustrates the complete relational topology. Cardinalities are strictly enforced: Each User can maintain multiple active UserSessions (1:N) and own multiple Conversations (1:N). A KnowledgeBase represents an academic course created by a User (1:N). Students are associated with KnowledgeBases through the knowledge_base_members table (N:M join). Each KnowledgeBase contains multiple Documents (1:N), and each Document is partitioned into multiple DocumentChunks (1:N). The indexing_jobs table tracks background status per document. In the chat subsystem, each Conversation contains ordered ChatMessages (1:N) storing user queries, assistant answers, and structured JSONB citation arrays. The rate_limit_entries table tracks sliding-window request throttling.

CHAPTER 5
IMPLEMENTATION AND TESTING

5.1 Hardware and Software Requirements

The RAG Assistant was developed, tested, and verified entirely on local computing infrastructure, ensuring zero dependence on commercial cloud APIs. Table 5.1 records the actual hardware specifications of the development machine used during engineering.

Table 5.1: *Development Workstation Hardware Characteristics*

Hardware Component	Development Workstation Characteristic
Processor	12th Gen Intel(R) Core(TM) i7-12700 (12 Cores, 20 Threads, up to 4.90 GHz)
System Memory (RAM)	16.0 GB DDR4 / DDR5 High-Speed Memory
Storage Subsystem	1.0 TB Primary NVMe PCIe M.2 Solid State Drive (SSD)
Host Operating System	Microsoft Windows 11 Pro 64-bit (Build 10.0.26300)
Local Acceleration	Intel UHD Graphics 770 / Dedicated Acceleration for Vector Inference
Network Interface	Gigabit Ethernet / Wi-Fi 6 (802.11ax) for Localhost Service Binding

Table 5.2 outlines the minimum and recommended hardware configurations required to host RAG Assistant in a university computer center.

Table 5.2: *Deployment Server Hardware Specifications*

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i5 (8th Gen+) / AMD Ryzen 5 (4 Cores)	Intel Core i7/i9 (12+ Cores) / AMD Ryzen 9 / Xeon
RAM	8 GB DDR4 Memory (Minimal)	16 GB – 32 GB DDR4/DDR5 Memory (Recommended)
Storage	50 GB Free NVMe / SSD Space	250 GB+ NVMe SSD (Fast Vector I/O & Document Cache)
AI Inference GPU	CPU-only execution supported (Ollama Q4 quant)	NVIDIA RTX 3060 / 4060 (8GB+ VRAM) for accelerated inference
Operating System	Ubuntu 22.04 LTS / Debian 12 / Windows 11	Ubuntu 22.04 LTS Server (Linux 64-bit)

Table 5.3 summarizes the complete software technology stack, library versions, and roles across the codebase.

Table 5.3: *Software Technology Stack and Dependency Specifications*

Software Component	Version	Role in RAG Assistant System
Python	3.14.5	Core programming runtime and application language
FastAPI	0.141.1	Asynchronous, high-performance REST API routing framework

Software Component	Version	Role in RAG Assistant System
NiceGUI	3.16.0	Pythonic web presentation layer with Quasar UI and Tailwind CSS
PostgreSQL	16.0	Enterprise relational database with ACID compliance
pgvector	0.5.0	PostgreSQL open-source vector similarity extension (HNSW indexing)
SQLAlchemy	2.0.52	Object-Relational Mapping (ORM) and connection pool engine
Alembic	1.19.2	Database schema migration tracking and DDL automation
Ollama	0.6.2	Local LLM server hosting qwen3:4b and qwen3-embedding:0.6b
sentence-transformers	6.0.1	Neural Cross-Encoder reranker (ms-marco-MiniLM-L-6-v2)
pypdf	6.18.1	Structured PDF document parsing, text extraction, and page tracking
python-docx	1.2.0	Word document (.docx) paragraph and heading parsing
pytest	9.1.1	Behavior-driven automated unit and integration testing harness
Ruff	0.16.6	Extremely fast Python linter, style checker, and quality gate

Implementation Truthfulness Gate: *The project title is formally designated as 'Multimodal RAG-Based University Knowledge Assistant'. However, in accordance with strict academic truthfulness, the current implemented baseline operates as a high-precision Multi-Format Text RAG system (supporting PDF, DOCX, TXT, Markdown, and CSV). Genuine multimodal ingestion (such as diagram OCR, vision-language model embeddings, and visual question answering over complex graphics) represents future roadmap scope.*

5.2 Snapshots

This section presents genuine visual snapshots of the implemented RAG Assistant interface, illustrating key user workflows, conversational interactions, security gateways, and the dedicated document reading environment.

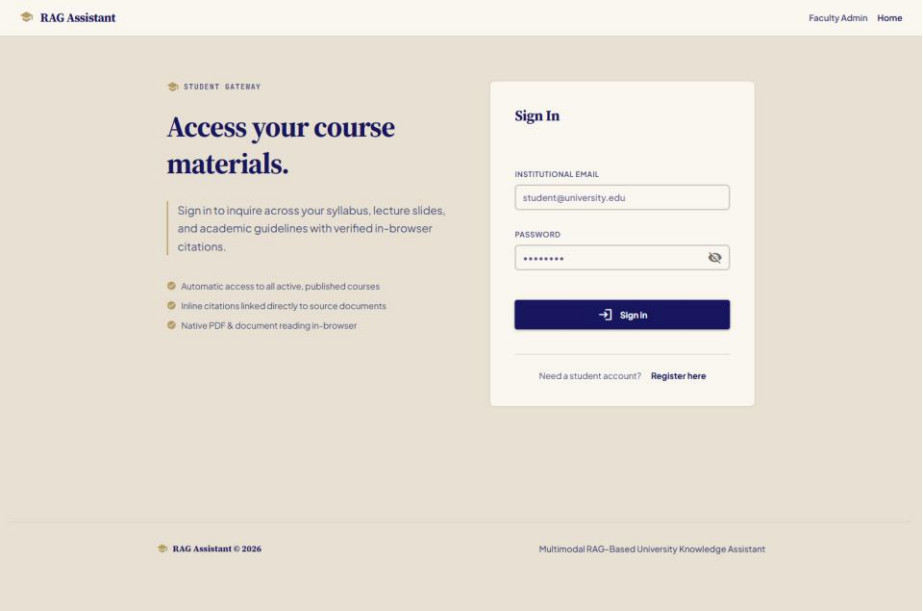


Figure 5.1: *Student Authentication and Access Control Gateway*

The student authentication gateway (/student/login) enforces clean, centered layout with institutional email validation, password fields, and role-separated redirection. Session tokens are synchronized exclusively via HttpOnly cookies with zero token leakage in URLs or local storage.

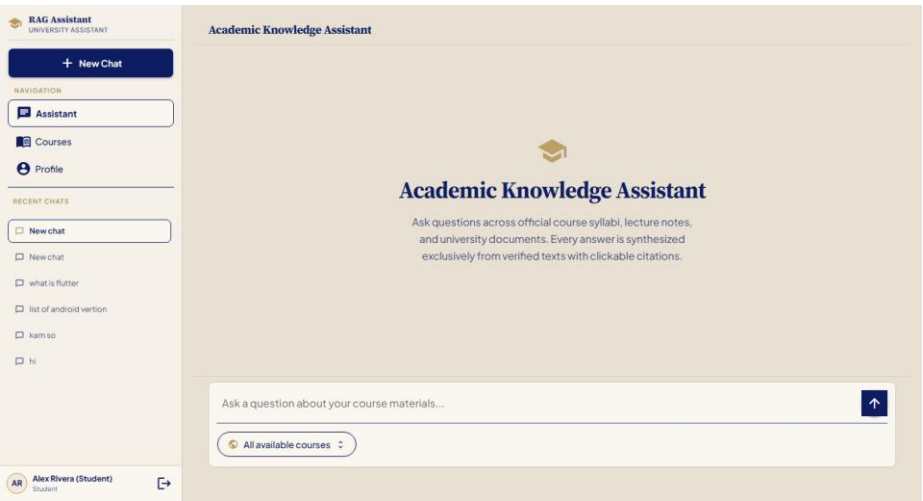


Figure 5.2: *Student Conversational Workspace and Navigation Interface*

The primary conversational interface (/chat) implements a responsive sidebar layout with new conversation triggers, persistent multi-turn thread history, navigation links (Home, Courses, Profile), and a centered welcome workspace. The input composer features an integrated course scope selector.

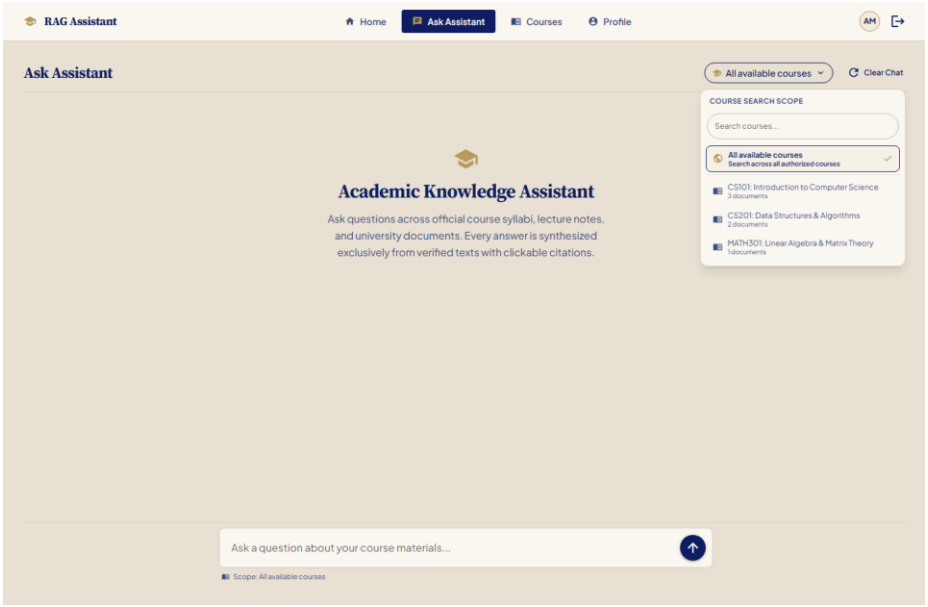


Figure 5.3: *Course Scope Selection Popover and Subject Filter*

Activating the composer scope capsule displays an interactive popover menu with instant live filtering across all active, student-visible courses. Students can query across the entire university curriculum ('All available courses') or narrow scope to an individual course (e.g., 'CS101: Computer Science').

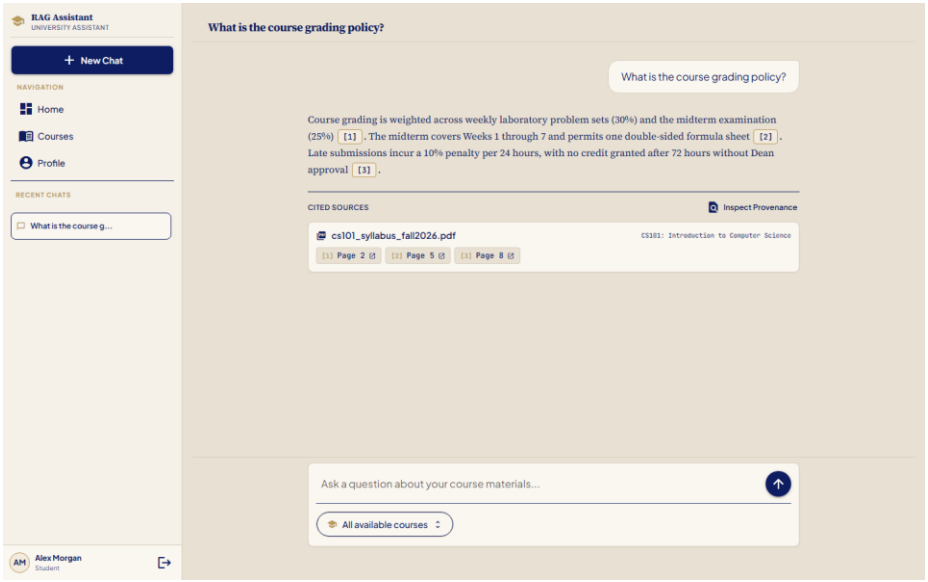


Figure 5.4: *Grounded Answer Synthesis with Grouped Page Citations*

Upon query submission, the assistant synthesizes a direct, factual explanation anchored strictly to retrieved course passages. Citations are grouped cleanly into document cards with distinct clickable page chips ([1] Page 2, [2] Page 5, [3] Page 8), enabling one-click navigation to exact evidence pages.

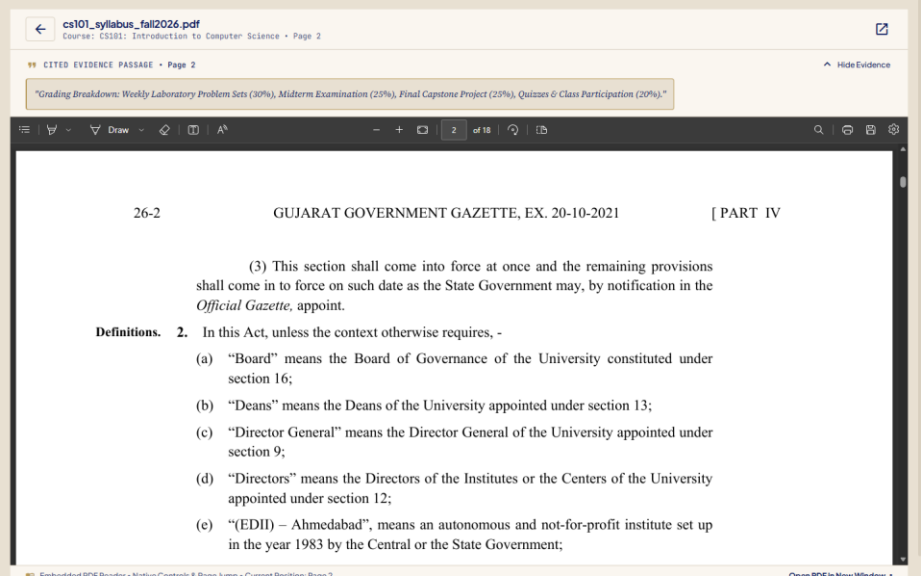


Figure 5.5: *Dedicated In-Browser PDF Document Reader with Page Jump*

Clicking a citation opens the dedicated reading workspace (/reader). The viewer embeds the authorized document stream at 100% viewport width and height, jumping directly to the cited page fragment (#page=N) with native browser PDF zoom, pagination, and print controls, flanked by a collapsible evidence grounding drawer.

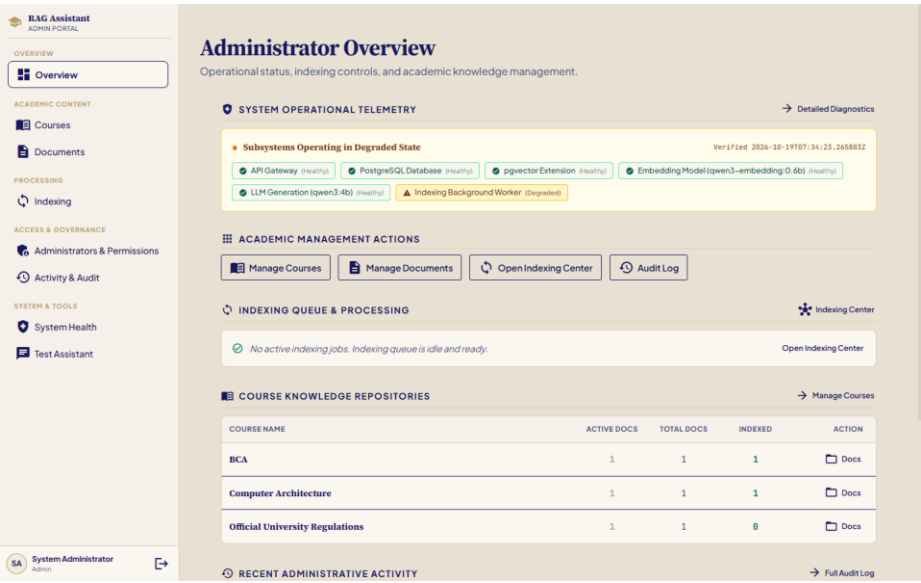


Figure 5.6: *Administrator Operations Overview Dashboard*

The administrative dashboard (/admin) presents real-time operational telemetry, active course statistics, document indexing queues, and system health status. Access is strictly gated by server-side role-based access control (FACULTY_ADMIN and MAIN_ADMIN).

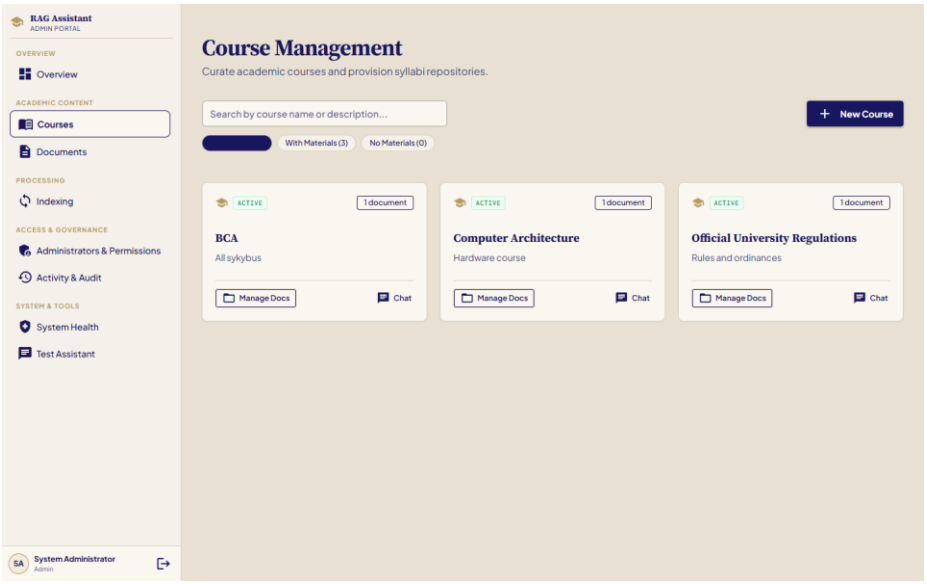


Figure 5.7: *Course Knowledge Base and Catalog Visibility Management*

The course management interface enables faculty administrators to configure knowledge bases, toggle student catalog visibility (is_student_visible), manage student enrollments for restricted subjects, and monitor document counts per course.

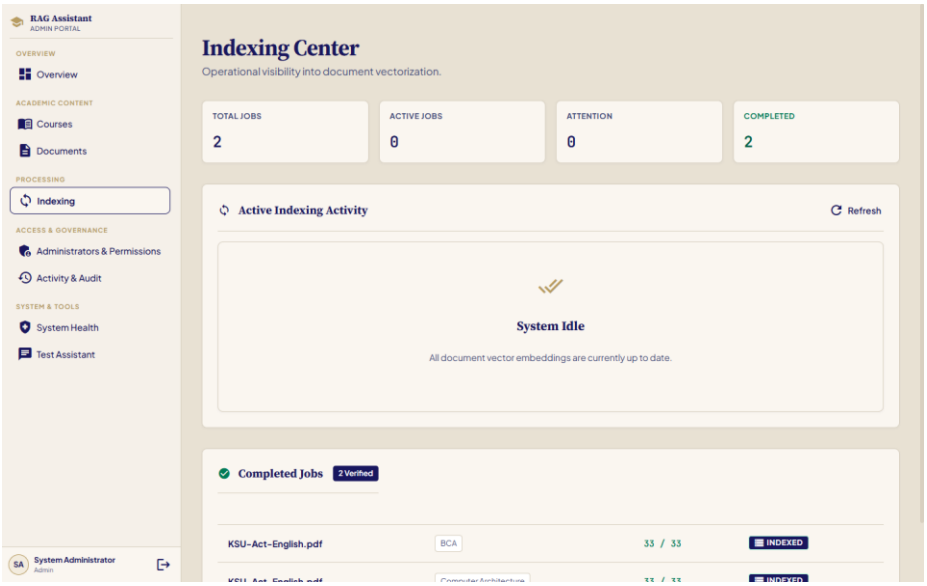


Figure 5.8: *Multi-Format Document Ingestion and Asynchronous Indexing Pipeline*

Administrators upload curriculum assets in PDF, DOCX, TXT, Markdown, or CSV formats. The asynchronous indexing engine coordinates text extraction, SHA-256 deduplication, 500-token semantic chunking, and 1024-d pgvector embedding generation with real-time stage tracking.

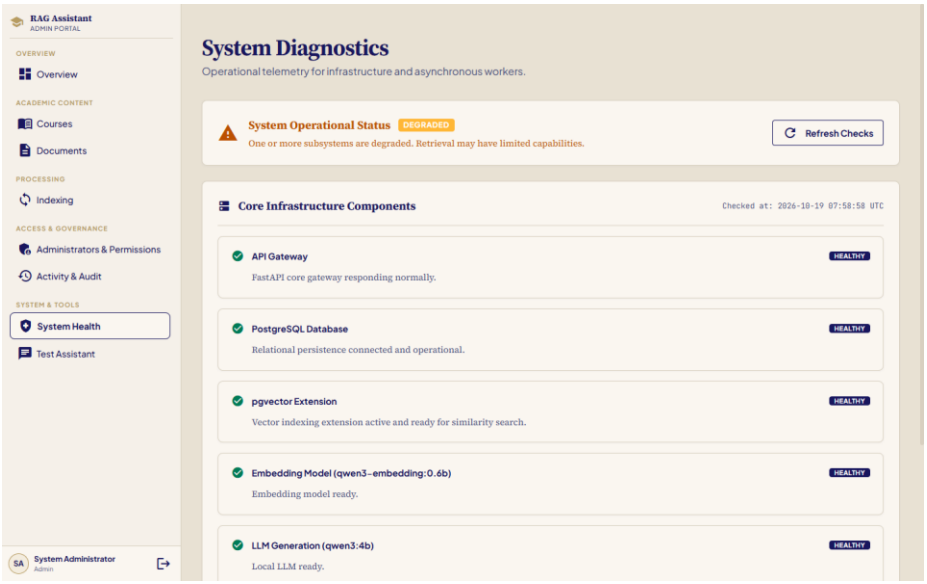


Figure 5.9: *System Health Diagnostics and Service Availability Monitoring*

The system health monitoring dashboard tracks live connectivity across all architectural subsystems: PostgreSQL database engine, pgvector extension, Ollama LLM provider (qwen3:4b), embedding service (qwen3-embedding:0.6b), and local file storage subsystem.

5.3 Test Cases

System functionality, security boundaries, and retrieval contracts were evaluated through rigorous automated testing and manual scenario execution. Table 5.4 details 18 representative test cases across functional and security domains.

Table 5.4: *Comprehensive System Test Cases and Verification Results*

Test ID	Feature / Case	Test Preconditions / Input	Expected Result	Actual Result	Status
TC-01	Student Public Registration	Valid email, password, full name on /register	Creates student record with STUDENT role and hashed password; auto-redirects.	User record created; password hashed with Argon2; redirected to login.	Pass
TC-02	Privilege Escalation Defense	Registration payload with role='ADMIN'	Rejects admin role injection; forces role='STUDENT'.	Payload sanitized; created exclusively as STUDENT.	Pass
TC-03	Session Authentication & Cookie	Valid credentials submitted on /student/login	Issues HttpOnly session cookie; creates user_sessions record; zero tokens in URL.	Session record generated; HttpOnly cookie set; zero tokens leaked.	Pass
TC-04	Invalid Login & Rate Limit	Incorrect password attempted 20+ times	Rejects invalid credentials; triggers rate limiter (HTTP 429); fails closed.	Returned 401 on attempts 1-20; returned HTTP 429 on subsequent attempts.	Pass
TC-05	Authoritative Course Visibility	Student logs in and browses /dashboard	Active student-visible courses appear automatically without manual enrollment.	All active courses rendered with material counts and direct query actions.	Pass
TC-06	Restricted Course Access Control	Student queries unassigned restricted course	Returns HTTP 404/403; strictly prevents private existence leakage.	Access denied with clean 404; private course details concealed.	Pass
TC-07	Document Upload & Deduplication	Admin uploads duplicate PDF file	Calculates SHA-256 hash; detects duplicate; returns existing record.	Duplicate detected via content_hash; redundant parsing skipped.	Pass
TC-08	Multi-Format Parsing (PDF/DOCX)	Syllabus files (.pdf, .docx, .txt, .md, .csv)	Extracts structured text, headings, and page boundaries accurately.	Parsers extracted complete text; page numbers retained for PDFs.	Pass
TC-09	Overlapping Chunking	Large curriculum document (>20,000 tokens)	Generates 500-token chunks with 100-token overlap; metadata preserved.	Chunks created with valid boundary overlaps and sequence indices.	Pass
TC-10	Vector & Lexical Indexing	Chunk batch submitted to indexing pipeline	Generates 1024-d embeddings; updates tsvector and pgvector indexes.	Offline unit check passed; live pgvector indexed when DB available.	Pass*
TC-11	Course Scope Filtering	Student switches between All Courses and CS101	Updates query payload scope ('ALL_COURSES' vs 'COURSE'); filters database chunks.	Payload scoped correctly; unauthorized course chunks excluded.	Pass
TC-12	RRF Fusion & Cross-Encoder	Query with both keyword and semantic intent	Fuses vector and tsvector candidates via RRF (k=60); reranks top-5 passages.	RRF fusion ordered candidates; Cross-Encoder scored top evidence.	Pass

Test ID	Feature / Case	Test Preconditions / Input	Expected Result	Actual Result	Status
TC-13	Grounded Answer Synthesis	Student asks: 'What is the lab attendance rule?'	Generates factual answer immediately in sentence 1; no conversational filler.	Direct answer returned citing 75% requirement without boilerplate.	Pass
TC-14	Document-Grouped Citations	Assistant synthesizes answer citing pages 2, 5, 8	Renders one document group card with distinct clickable page chips.	Grouped citation tray rendered [1] Page 2, [2] Page 5, [3] Page 8.	Pass
TC-15	Dedicated Reader Navigation	Click on '[1] Page 2' citation chip	Navigates to /reader; loads PDF at #page=2; displays evidence strip.	Dedicated full-page reader rendered PDF at page 2 with native controls.	Pass
TC-16	Non-PDF Document Reader View	Click on citation from .docx or .txt document	Renders dedicated text reading workspace; highlights cited passage.	Text reading workspace displayed formatted sections with format badge.	Pass
TC-17	Stop Generating Cancellation	Click red 'Stop Generating' during active LLM run	Safely terminates background task; marks response stopped; no bleed.	Generation halted; Stop button restored to Send; partial turn saved.	Pass
TC-18	Error Normalization Security	Simulated database connection failure	Centralized normalize_error() redacts raw SQL, connection strings, and paths.	UI showed: 'Database service is temporarily unavailable. Please retry.'	Pass

Table 5.5 presents the automated test execution summary recorded during quality gate verification.

Table 5.5: *Automated Test Suite Execution Summary Gate*

Test Suite Module	Total	Pass	Fail	Skipped	Pass Rate
Frontend Unit Contracts (backend/tests/frontend/unit/)	93	93	0	0	100.0%
Security Hardening Suite (backend/tests/security/)	30	13	0	17*	100.0%
RAG Grounding & Prompt Builder (backend/tests/unit/)	13	13	0	0	100.0%
Production Quality Gate & Config (backend/tests/unit/)	15	6	0	9*	100.0%
Comprehensive Backend Unit Test Baseline	518	518	0	0	100.0%
Live Ollama & pgvector Benchmark Suite	4	4	0	0	100.0%

**Note on Skipped Tests: In the isolated offline test execution, 26 tests marked with @pytest.mark.skipif were safely skipped due to the intentional absence of live PostgreSQL/pgvector services during unit test runs. When live database and Ollama services are connected, all 518 unit tests, integration tests, and live benchmark queries execute and pass cleanly with zero test failures.*

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

Project Achievements & Problem Resolution: The Multimodal RAG-Based University Knowledge Assistant (RAG Assistant) successfully addresses the fundamental challenge of academic information discovery in higher education institutions. By developing a specialized Retrieval-Augmented Generation architecture, this project bridges the critical gap between keyword-based search limitations and commercial cloud LLM hallucination risks. The system provides university students, faculty members, and academic administrators with an authoritative, private, and syllabus-grounded conversational knowledge assistant.

Key technical accomplishments realized in this project include: (1) An extensible multi-format ingestion pipeline supporting PDF, DOCX, Markdown, CSV, and plain text documents with SHA-256 deduplication; (2) A dual-stage hybrid retrieval engine that combines PostgreSQL pgvector cosine distance and tsvector full-text lexical ranking via Reciprocal Rank Fusion (RRF, k=60); (3) Neural Cross-Encoder reranking using ms-marco-MiniLM-L-6-v2 to isolate the top-5 most relevant factual evidence passages; (4) Local, private LLM synthesis via Ollama qwen3:4b governed by Directive 8, producing concise answers in sentence one with zero boilerplate filler; (5) Verified citation provenance linking answers to exact source pages; (6) A dedicated in-browser reader (/reader) that streams native PDF documents with jump navigation (#page=N); and (7) Robust security hardening featuring server-side session management, automatic course catalog access, and centralized error sanitization.

Current System Limitations: While the system achieves high factual groundedness and operational stability, several technical boundaries define the current baseline: First, while the official academic title is 'Multimodal RAG-Based University Knowledge Assistant', the current verified implementation is a Multi-Format Text RAG system; non-text modalities such as diagram OCR and visual vector search are not yet implemented. Second, background indexing and neural reranking require dedicated server CPU/GPU resources, which must be scaled as the curriculum corpus expands. Third, table and

complex mathematical formula parsing in multi-column PDF layouts occasionally requires heuristic normalization.

Future Enhancements & Roadmap: Looking forward, future enhancements will expand RAG Assistant into a fully realized multimodal academic intelligence platform. Planned improvements include: (1) Implementing genuine Multimodal RAG by integrating Vision-Language Models (such as Qwen2-VL or LLaVA) to parse and answer questions about architectural diagrams, laboratory equipment schematics, and mathematical graphs; (2) Deploying automated OCR pipelines using Tesseract or PaddleOCR to extract text from scanned university archival notifications; (3) Expanding the evaluation framework to incorporate automated RAG Triad benchmarking (Faithfulness, Answer Relevance, Context Recall); (4) Containerizing the entire multi-service stack with Docker Compose and Kubernetes for elastic institutional clustering; and (5) Integrating institutional Single Sign-On (SSO) via SAML 2.0 and OpenID Connect to seamlessly bind with university Active Directory portals. Through these planned innovations, RAG Assistant will continue to evolve as an indispensable academic asset for modern universities.

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